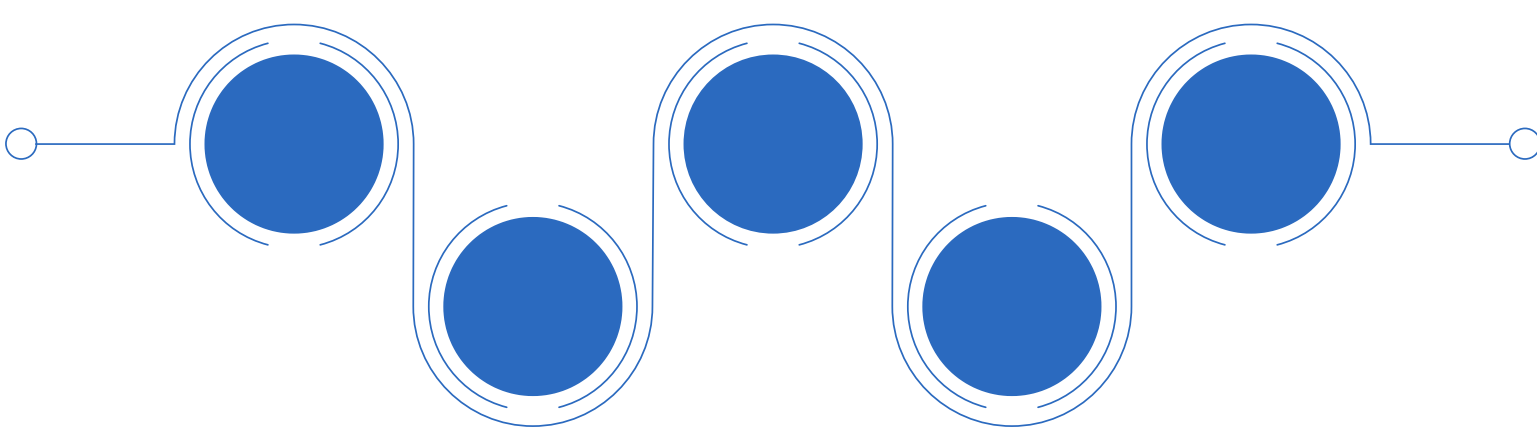


CS6120

NLP

Optimizing Retrieval-Augmented Generation for Construction Management: An Ablation Study on Embedding and Generative Models



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INTRODUCTION

- The construction industry's growing project complexity and strict safety regulations often overwhelm traditional manual monitoring and reporting methods.
- Massive volumes of technical specifications, regulations, and unstructured data impede swift, accurate decision-making.
- Integrating generative AI through RAG models promises automated retrieval and contextual interpretation of domain documents, helping reduce decision making times significantly.

OBJECTIVE

- Build a domain-specific RAG system to automate retrieval of relevant safety standards and technical guidelines.
- Conduct an ablation study on embedding and generative strategies to determine the most effective configuration.
- Evaluate generative model outputs using BLEU and ROUGE metrics against a curated ground-truth from Wendel specifications.

LITERATURE REVIEW

- **CONSTRUCTA Framework [1]:** Combines Static RAG, Knowledge RAG, and Reinforcement Learning from Human Feedback to automate construction scheduling through context-aware retrieval and expert-aligned planning.
- **Entropy-Optimized Dynamic Text Segmentation (EDTS) [2]:** Improves traditional retrieval by dynamically chunking construction documents into semantically coherent sections using entropy calculations for better information clarity.
- **AEC Q&A Advisor [3]:** A RAG-based system reduces LLM hallucinations in technical document QA by combining advanced prompt design (e.g., CoT, Few-shot) with cosine similarity-based retrieval, outperforming traditional search tools.

[1] Yifan Zhang, Xue Yang (2025) CONSTRUCTA: Automating Commercial Construction Schedules in Fabrication Facilities with Large Language Models. arXiv:2502.12066v1 [cs.AI] 17 Feb 2025

[2] Haiyuan Wang, et al. (2025) Entropy-Optimized Dynamic Text Segmentation and RAG-Enhanced LLMs for Construction Engineering Knowledge Base. doi.org/10.3390/

[3] Yunshun Zhong, Tamer El-Diraby (2025) Generative project question answering system: triangulating three approaches for project authoring

LITERATURE REVIEW

- **Safety Knowledge Retrieval with RAG vs. Fine-Tuned LLM [4]:** A comparative study showing RAG and fine-tuned models outperform GPT-4 in construction safety tasks, improving BLEU/ROUGE scores and reducing hallucinations using domain-specific prompts and embeddings.
- **RAG4CM System [5]:** Introduces a knowledge graph-based RAG pipeline that chunks multimodal documents semantically and incorporates user-preference learning, achieving high accuracy in construction management information retrieval.

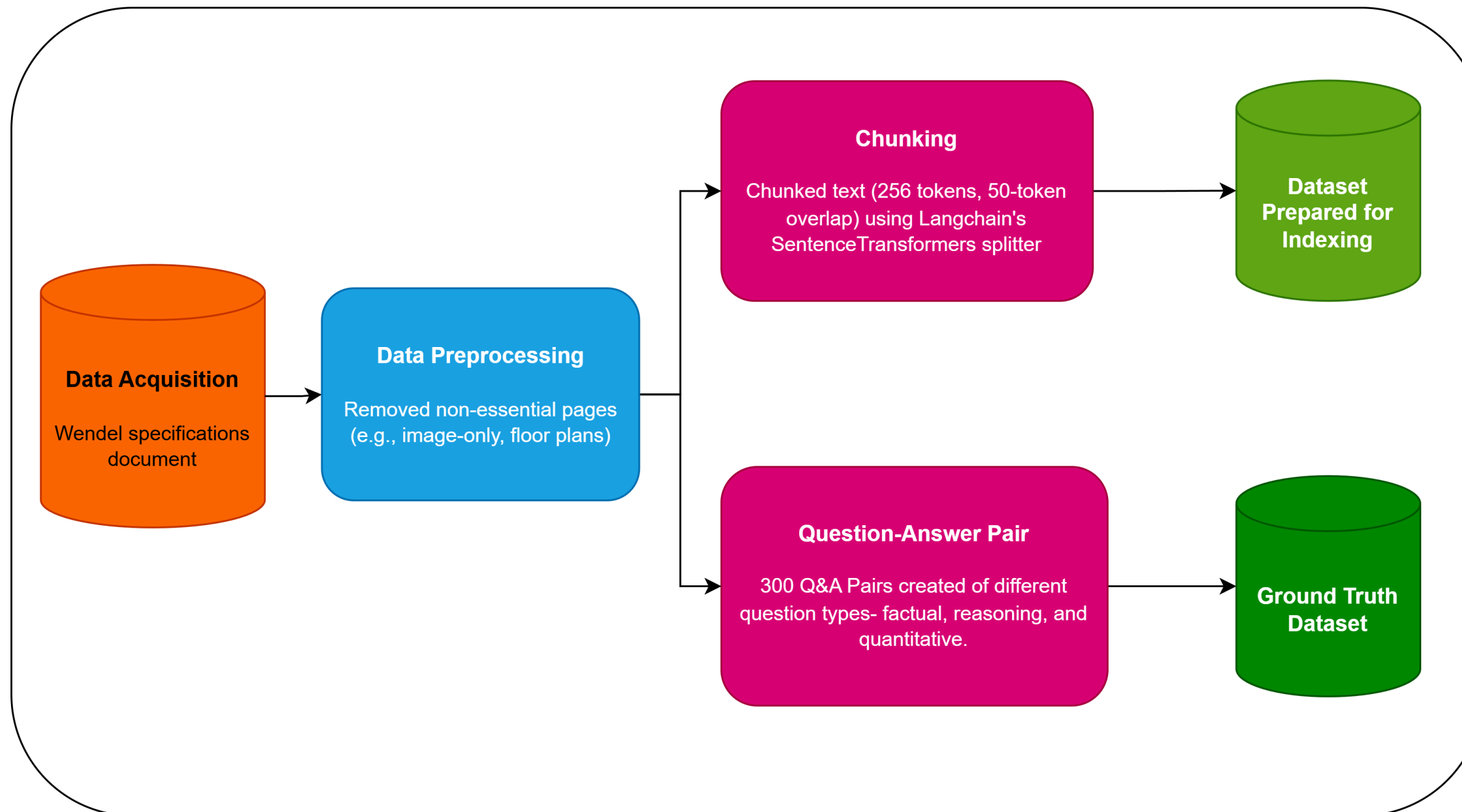
[4] Jungwon Lee et al. (2024) Performance comparison of retrieval-augmented generation and fine-tuned large language models for construction safety management knowledge retrieval doi.org/10.1016/j.autcon.2024.105846

[5] Chengke Wu et al. (2025) Retrieval augmented generation-driven information retrieval and question answering in construction management doi.org/10.1016/j.aei.2025.103158

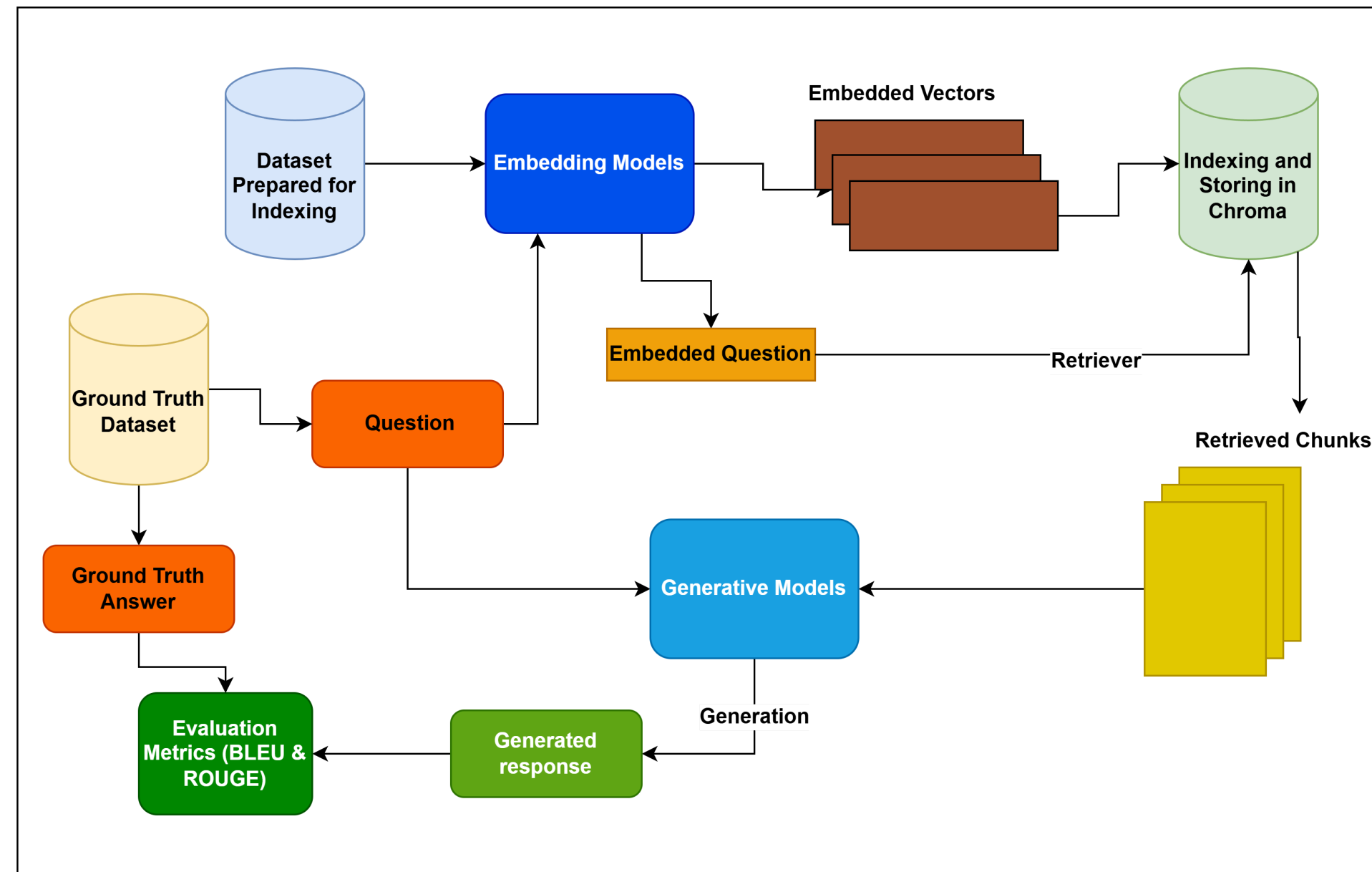
DATASET DESCRIPTION

- Utilized the Wendel specifications document on office building renovation, the dataset captures detailed construction processes, codes, and safety regulations.
- The document's consistent formatting and clear sectioning helps segment content and align each Q&A pair with its relevant section.
- Organized as a CSV with columns for Question Type, Question, Answer, and Source Section, featuring factual, reasoning, and quantitative questions for comprehensive domain coverage.

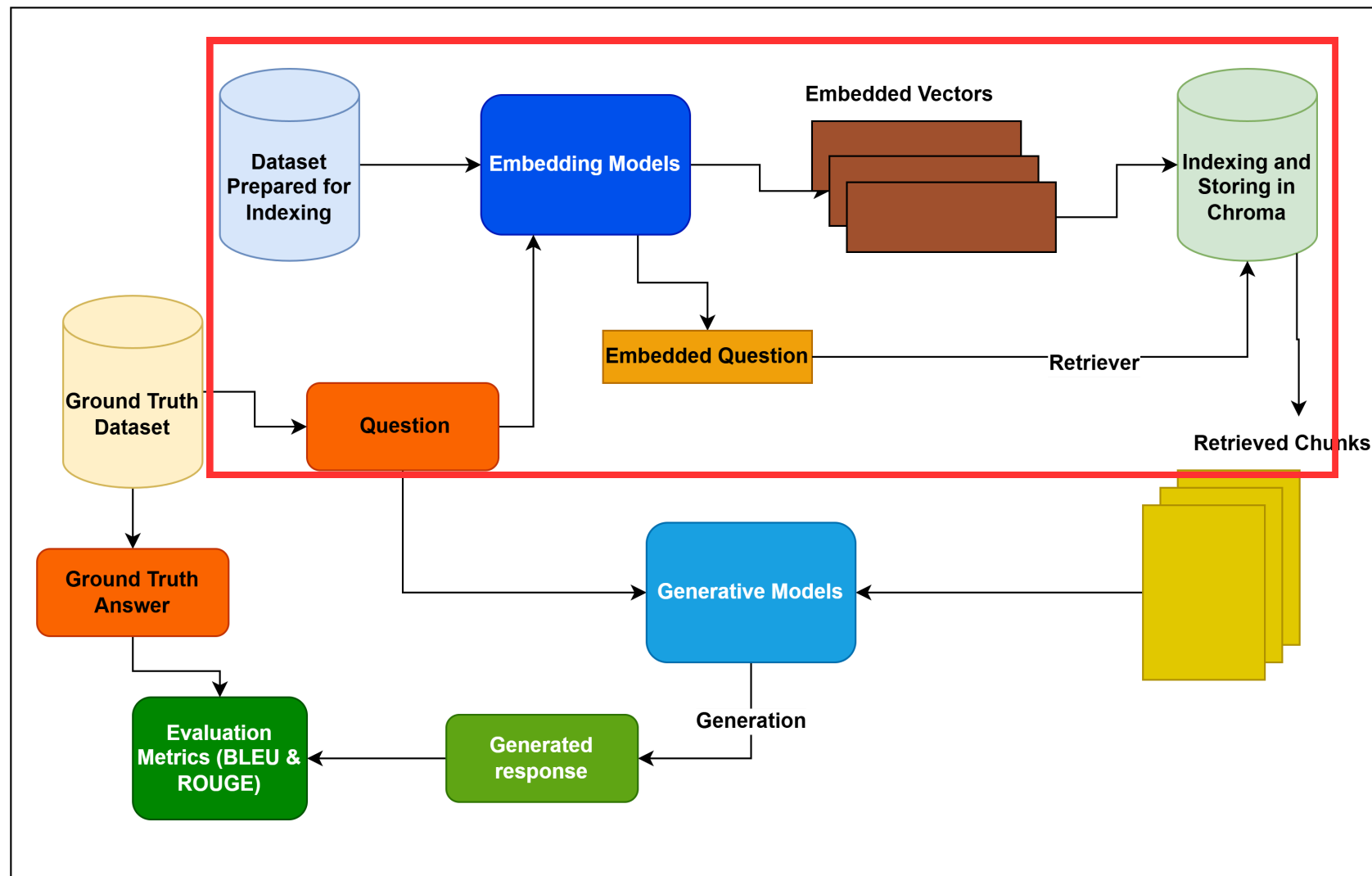
DATA PREPROCESSING & GROUND TRUTH GENERATION



EXPERIMENTAL METHODOLOGY

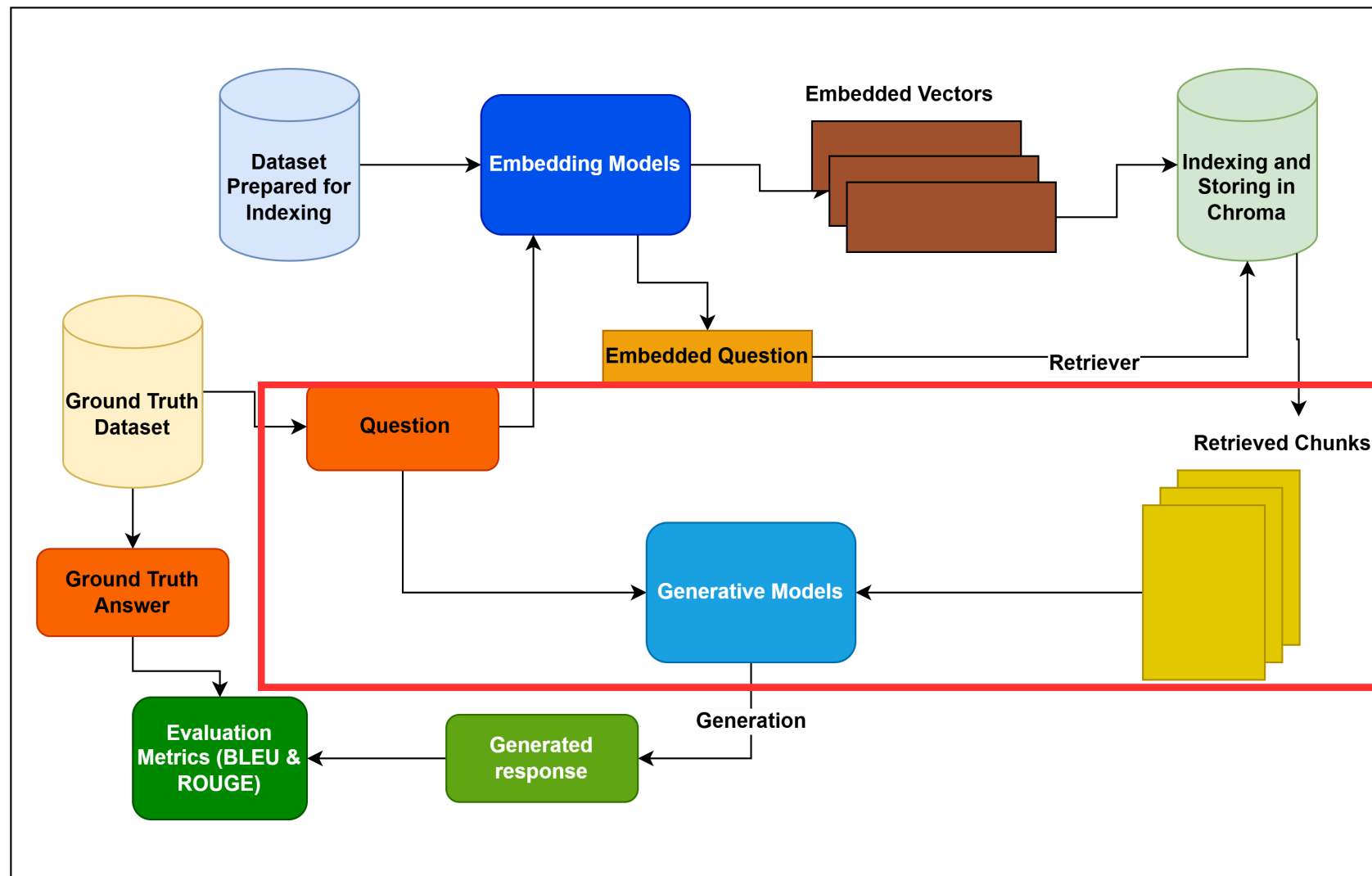


EMBEDDING PIPELINE



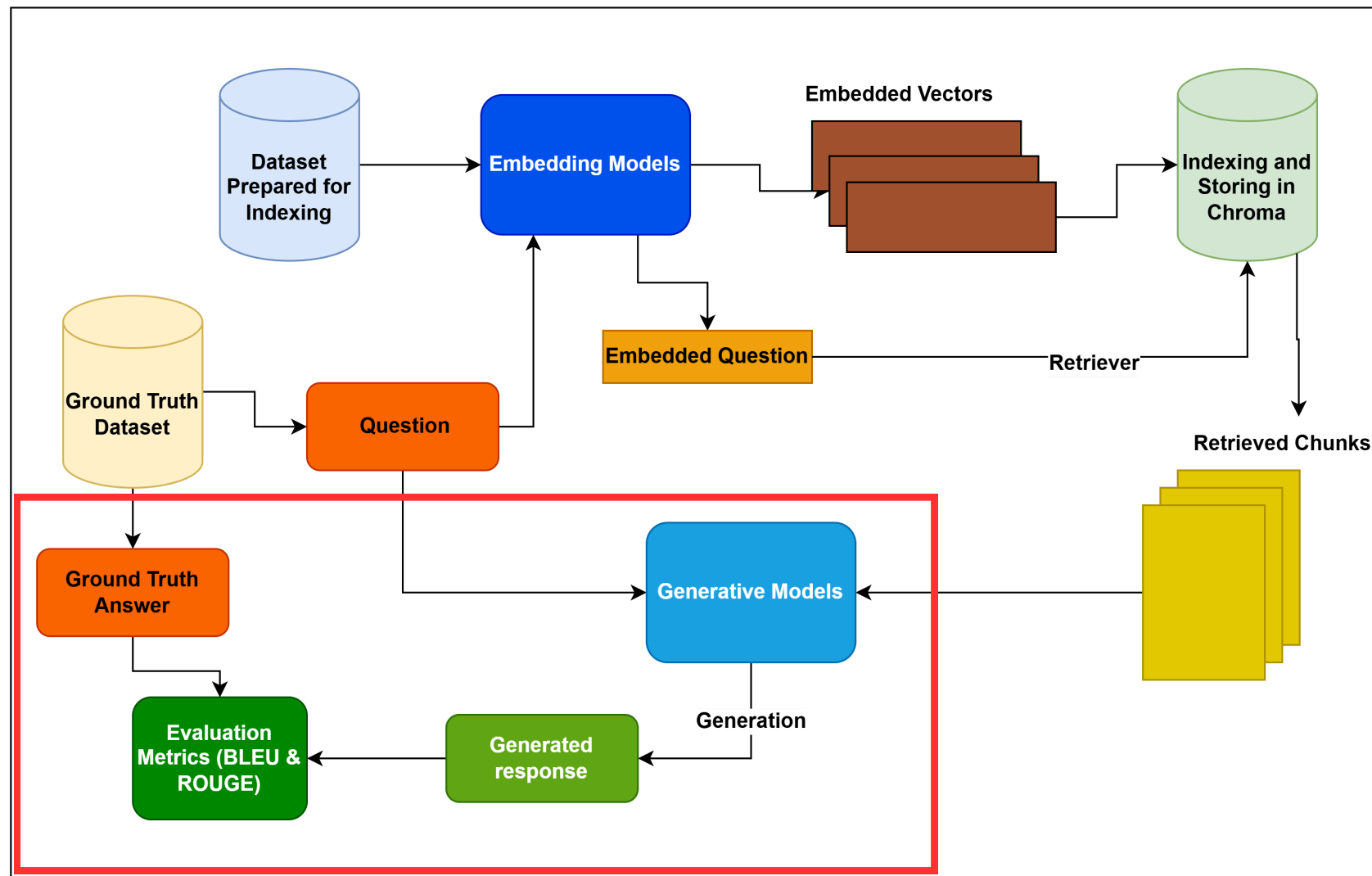
- **IBM Granite Embedding 278M:** Utilizes a BERT-based biencoder with 384-dimensional vectors, 6 blocks, 512-token context, and CLS pooling—designed for efficient, text-only encoding of standard-length inputs.
- **nomic-embed-text:latest:** Built on a “nomic-bert” architecture, it supports longer contexts (up to 2048 tokens) with 12 blocks, 768-dimensional embeddings, 12 attention heads, and mean pooling, excelling in handling complex, lengthy passages.

GENERATION PIPELINE



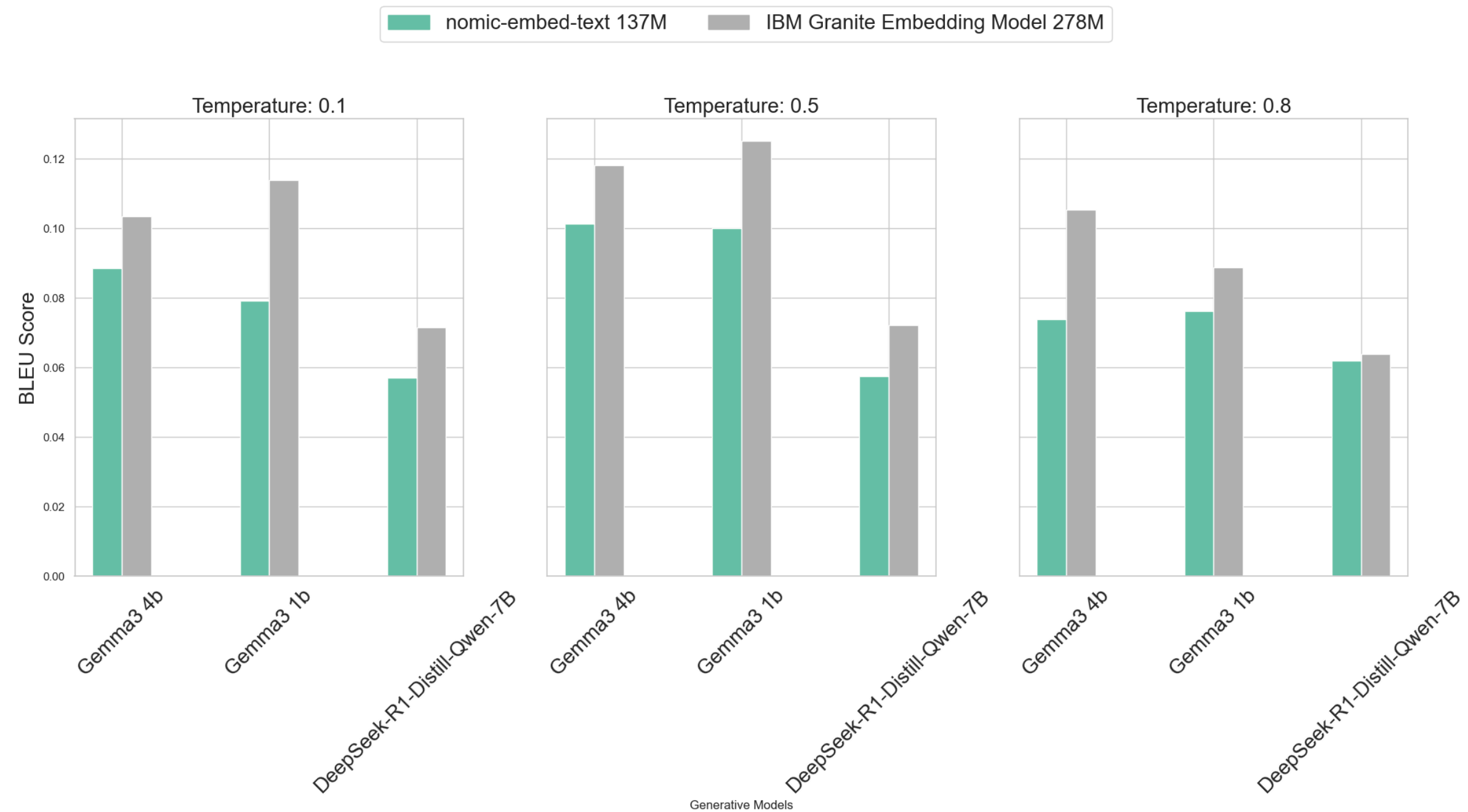
- **Gemma Models:** Evaluated gemma3:1b (1B params, 32K token context, BERT-inspired, resource-efficient) and gemma3:4b (4B params, 128K token context, multimodal, extended context).
- **DeepSeek Model:** Assessed deepseek-r1:7b (7B params, 131K token window, 28 attention heads) that distills reasoning from larger architectures.

EVALUATION PIPELINE

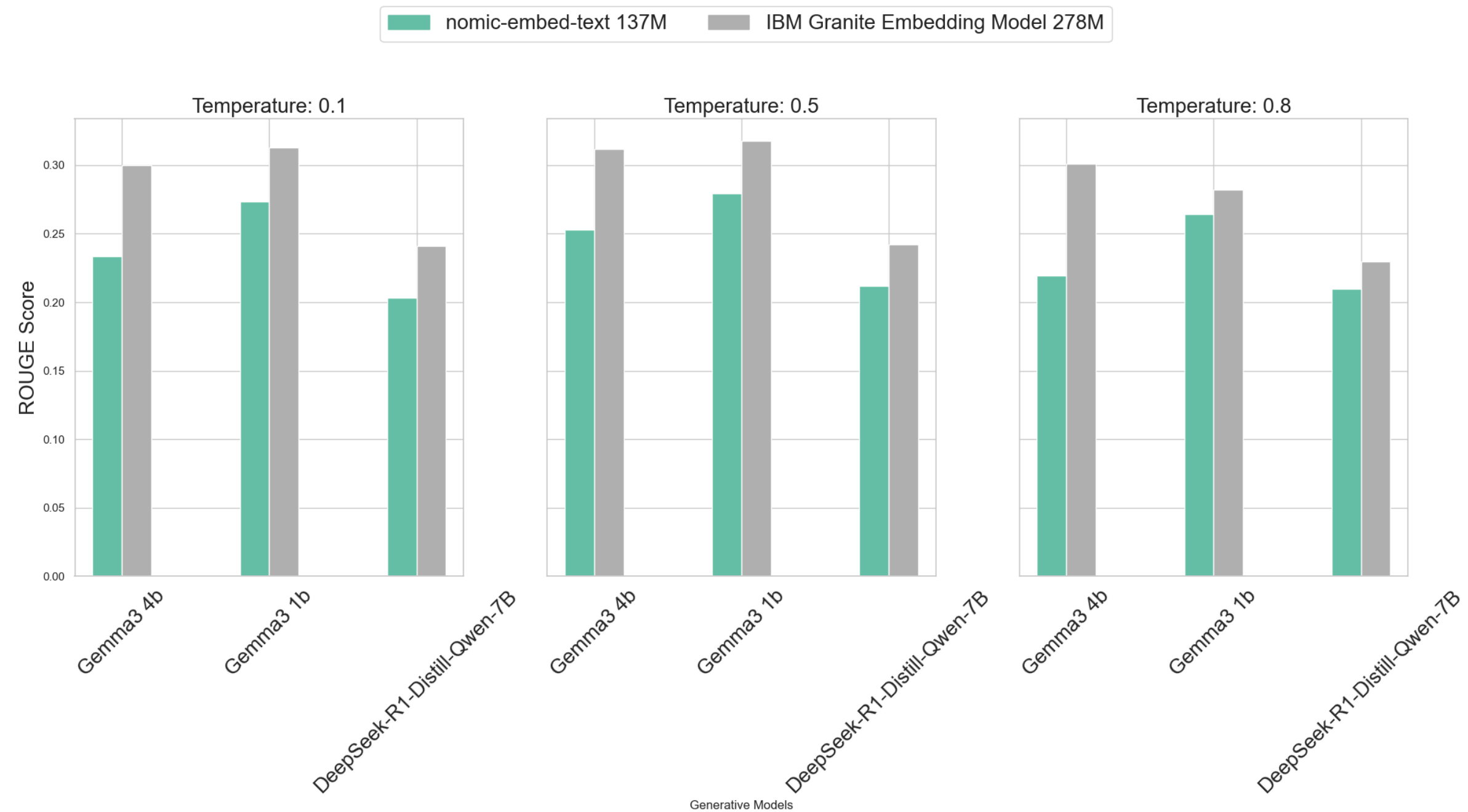


- **BLEU Score:** It measures precision by evaluating how much of the generated text overlaps with the reference answer's wording and structure (n-gram match).
- **ROUGE Score:** It complements this by measuring recall, assessing how much of the reference content is captured in the generated output (including ROUGE-N and ROUGE-L variants).

BLEU SCORES

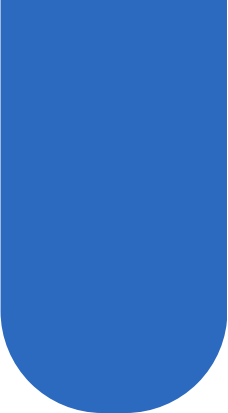


ROUGE SCORES



CONCLUSION

- **Best Performance Configuration:** The IBM Granite Embedding Model (278M) combined with Gemma3 1b at a 0.5 temperature achieved the highest BLEU (0.1252) and ROUGE (0.3178) scores. Its BERT-based 384-dimensional embeddings captured semantics in construction texts better than nomic-embed-text 137M. Notably, the lighter Gemma3 1b sometimes outperformed larger models, offering focused and less noisy responses.
- **Impact of Temperature and Model Design:** Temperature tuning was critical—0.5 struck the ideal balance between determinism and diversity. Lower values (0.1) led to overly rigid outputs; higher values (0.8) reduced precision and recall.



THANK YOU